

invention_oevaluat[®]r

OBJECTIVE IDEA ANALYSIS

Invention Name: Electromagnetic Inductive Suspension and Stabilization System for a Ground Vehicle

Submitted By: User-provided patent PDF

Invention ID: US 3,470,828

Date of Report: 2026-08-16

This report was generated by the inventionevaluator report-automation platform from the Invention Evaluation Engine v1.6 pipeline (evidence sufficiency architecture). It compiles established findings, analytical conclusions, and an operational audit produced by the nine-phase evaluation. Scores shown are derived executive summaries of the underlying findings — they are presentation-layer derivations, not new evidence, and do not constitute an overall patentability label. This document is not legal advice.

Preliminary evidence-constrained evaluation. Not legal advice, not a formal patentability opinion, and not an FTO opinion.

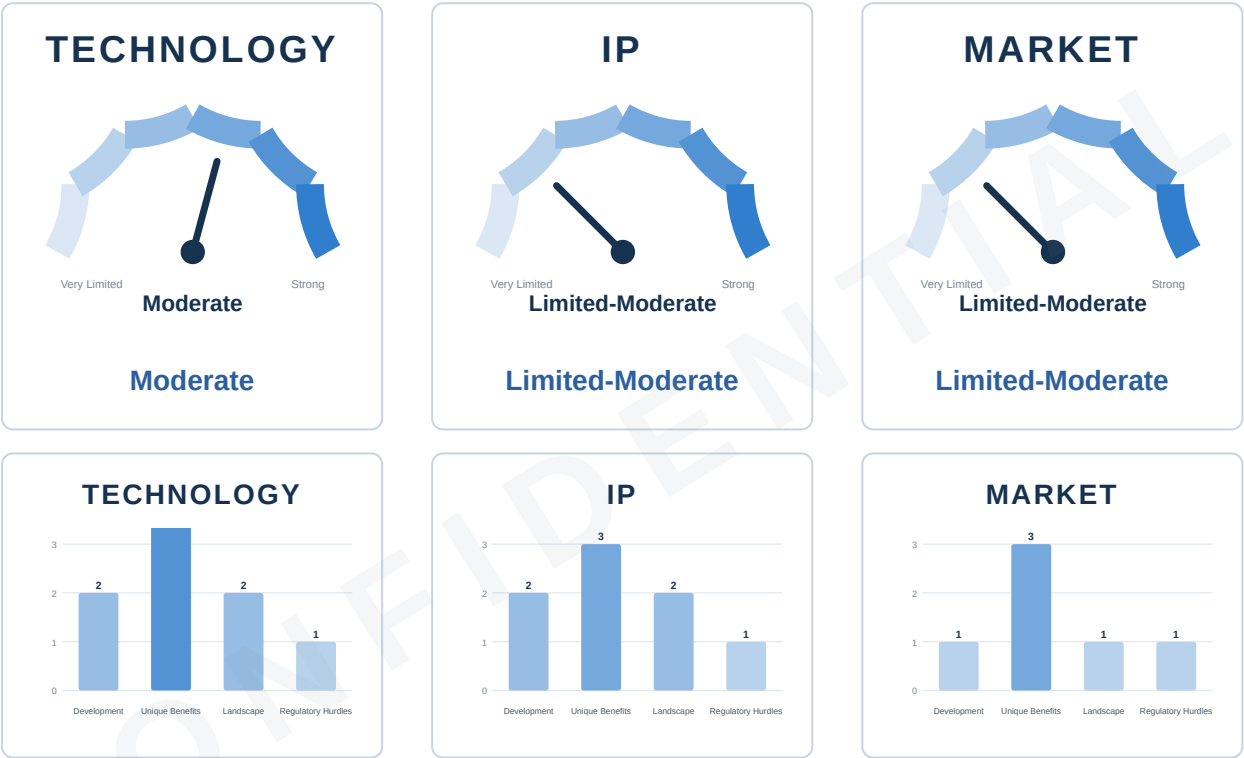
Table of Contents

Executive Summary	3
Technology Analysis	5
Patent Landscape Analysis	6
IP / Novelty Analysis	7
Literature Analysis	8
Market Analysis	9
Opportunity Assessment	10
Potential Partners	11
Decision Matrix	12
Operational Audit	13
Appendices	14
SWOT Analysis	15
Landscape & Market Data	16

Executive Summary

Derived executive summary — constrained to Established Findings + Analytical Conclusions (Executive Summary $\subseteq$ findings + conclusions). Gauge and bar scores are presentation-layer derivations from the findings; basis proposition_ids are listed in the scores manifest.

Very Limited Limited Limited-Moderate Moderate Moderate-Strong Strong



Commercialization Summary

Commercial actionability is Indeterminate-to-Limited on the completed evidence. The patent discloses a technically specific passive electrodynamic suspension architecture, but market size, deployment history, current rights, and product-level adoption remain in the recovery queue.

- Strong historical technical disclosure with substantial infrastructure and cryogenic burden. [more](#)
- Strongest identified pathway is technical and historical-technology diligence, not an unqualified standalone patent license. [more](#)

Evidence trace

The proposition register below is extracted from the compiled report so every score basis can be checked against visible report content.

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Technology Analysis

1.1 Technology profile

See `technology-profile-us3470828.md`. The system is a passive electrodynamic suspension architecture combining superconducting vehicle magnets, conductive track-loop arrays, geometry-dependent restoring forces, long damping loops, and current-limiting inductors.

1.2 Technical evidence

The primary record directly supports the existence of the disclosed architecture and claims. The patent's theoretical power, train-size, speed, lift, and force figures are recorded as patent disclosure claims. They are not independently validated performance findings.

Patent Landscape Analysis

2.1 Landscape result

See patent-landscape-us3470828.md. Earlier magnetic levitation and suspension records were located, but a complete family-deduplicated quantitative landscape was not established.

2.2 Design-space position

The target patent explores a broad design space: vertical lift loops, figure-eight lateral loops, long damping loops, buried or road-level loops, alternate orientations, and grouped current-limiting circuits. This is evidence of a broad architecture exploration event, not by itself evidence of patentability strength.

IP / Novelty Analysis

3.1 Provisional evaluation claim

The claim representation is analytical and not a legal claim construction. It requires the vehicle, superconducting magnets, longitudinal shorted conductive loops, inductive relationship, speed-dependent lift, equilibrium position, and vertical stabilization.

3.2 Prior-art mapping

US1020943A and US3125964A disclose related magnetic vehicle concepts but differ in important ways. The retrieved evidence does not establish that either one alone discloses the complete superconducting passive shorted-loop architecture and vertical restoring relationship.

3.3 Anticipation gate

No anticipation finding is admitted. Claim-level search and mapping remain incomplete.

3.4 Obviousness and causal bridge

The bridge from earlier magnetic levitation to the target combination requires evidence for the specific use of superconducting vehicle magnets, passive longitudinal shorted-loop arrays, speed-dependent induced lift, and vertical equilibrium stabilization. Motivation, compatibility, reasonable expectation of success, and unexpected technical result are not established.

3.5 Chronology

The target filing/priority date is recorded as 1967-11-21 from the primary patent record. Earlier retrieved patent references include US1020943A (1911 priority) and US3125964A (1964 publication). A complete family and prosecution chronology remains in recovery.

Literature Analysis

The target patent names Powell's 1963 ASME paper, Van Beelen et al.'s 1965 *Physica* paper, and Powell/Danby's 1966 ASME report. Their citation identity is present in the patent, but full independent retrieval and proposition-level mapping were not completed. No independent literature disclosure is admitted as a final finding.

Market Analysis

The plausible market boundary includes maglev/high-speed rail infrastructure, superconducting magnet systems, conductive guideways, cryogenic support, and future low-pressure transport. No reconstructable market size, CAGR, shipment, revenue, adoption, or current product-identity finding was established.

Commercial barriers include guideway cost, cryogenic operation, quench and safety management, alignment, transitional-speed operation, infrastructure approvals, and reliability requirements.

Opportunity Assessment

The technology is technically distinctive at the patent-disclosure level but commercially constrained by infrastructure intensity and unresolved deployment evidence. Modern resurrection could be investigated using contemporary superconductors, sensing, control, power electronics, and materials, but that is an analytical opportunity hypothesis rather than an established market conclusion.

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Potential Partners

The strongest identified pathway is a technical consortium or research collaboration involving maglev system integration, superconducting magnets, cryogenics, guideway infrastructure, and transportation research. No organization-specific partner-fit finding is admitted because sell/buy/need/mapping fields were not independently verified.

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Decision Matrix

Gate	State	Basis
Utility	Evidence-supported at disclosure level	Patent describes a practical vehicle suspension objective and embodiments
Novelty / anticipation	Unresolved — search incomplete	Related references found; complete claim mapping and recovery remain open
Inventive step	Unresolved — bridge incomplete	Motivation, compatibility, expectation, and unexpected result not established
Commercial actionability	Indeterminate-to-Limited	Market, adoption, infrastructure, and rights evidence incomplete
Partner fit	Unresolved	Candidate categories identified; organization-specific evidence incomplete

Operational Audit

Every unresolved proposition is retained in `avenue-ledger-us3470828-v17.md` with recovery records. The current run does not convert retrieval difficulty, public-status labels, or missing market data into factual absence claims.

Key remaining evidence debt:

- Complete patent family, continuity, prosecution, and jurisdiction review.
- Execute five-class recovery for claim-level anticipation and obviousness.
- Independently retrieve and map the cited technical literature.
- Establish any physical prototype, test, pilot, or deployment evidence.
- Build a bounded and source-reconstructable modern market model.
- Verify commercial adoption and current rights/asset layers.
- Verify organization-specific partner fit.

Appendices

- Submission: submission-us3470828.md
- Technology profile: technology-profile-us3470828.md
- Patent landscape: patent-landscape-us3470828.md
- Novelty search: novelty-search-us3470828.md
- Literature search: literature-search-us3470828.md
- Market analysis: market-analysis-us3470828.md
- Partner analysis: partner-analysis-us3470828.md
- Avenue ledger: avenue-ledger-us3470828-v17.md
- Evidence graph: evidence-graph.json
- Constraint report: constraint-report.json
- Evidence debt: evidence-debt.json

SWOT Analysis

STRENGTHS

- Detailed passive suspension and stabilization architecture
- Multiple disclosed loop geometries

WEAKNESSES

- Cryogenic superconducting burden
- No independent performance validation established

OPPORTUNITIES

- Historical technology resurrection using modern materials and controls
- Research collaboration or infrastructure-subsystem pathway

THREATS

- Capital-intensive guideway deployment
- Safety, reliability, and regulatory burden

COMMERCIAL
ACTIONABILITY

Indeterminate-to-
Limited

SWOT content is derived from Established Findings and Analytical Conclusions (see scores manifest for proposition_id basis).

Landscape & Market Data

PATENT APPLICATIONS BY REGION

Data not established in this run

See Operational Audit — region split not established from completed evidence protocol.

PATENT APPLICATIONS BY MAIN APPLICANTS

Data not established in this run

See Operational Audit.

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PATENT ACTIVITY TREND

Yearly series not established in this run

No defensible yearly series was established in this run; see Operational Audit.

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INDUSTRY METRICS

Time series not established in this run

No invention-specific industry metrics were established in this run; see Operational Audit.

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ESTABLISHMENTS BY EMPLOYEE-SIZE BRACKET

Bracket data not established in this run

See Operational Audit.

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